

Ashkan Y. Zadeh

PhD Researcher | Applied AI

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About Me

Applied AI researcher building and evaluating Human-Centred & eXplainable AI solutions, currently focused on their applications for autonomous driving. Curiosity leads me, challenge shapes me, and impact drives me. Hi, I'm Ashkan.

Technical Skills

Machine Learning

PyTorch, fine-tuning, transfer learning, cross-modal alignment, benchmarking & evaluation

Generative AI & NLP

LLM reasoning pipelines, vision-language models (VLMs), prompt engineering, RAG, semantic search, information retrieval, computational linguistics, computational psycholinguistics

Multimodal Systems

Multi-sensor fusion & alignment, spatial grounding, autonomous driving frameworks

Human Factors & UX Research

User study design, field & questionnaire-based data collection, human-subject experiments, trust & explainability evaluation, inter-annotator agreement, mixed-methods (qualitative + quantitative) analysis

Software Engineering

Python, object-oriented design, REST APIs, testing, version control

Data & Statistics

ETL pipelines, exploratory data analysis, statistical hypothesis testing, relational & NoSQL databases

Selected Research Projects

NARRATE — Open Multimodal Driving Dataset (First Author, ECCV 2026)

[Code](#) | [Dataset DOI](#) | [OpenReview](#)

- Designed and ran a real-world field study collecting **2,050 multimodal driving events** from **35 drivers**, pairing on-road multi-sensor capture (camera, LiDAR, GNSS, IMU, telemetry) with participant questionnaires and span-level situational-awareness labels via **Label Studio**.
- Engineered **Python** ingestion pipelines for automated multi-sensor synchronization, cleaning, and participant-disjoint splitting, eliminating train/test leakage across drivers.
- Benchmarked transformer backbones (**RoBERTa**, **CLIP**, **T5**, **BERT**) on perception and anticipation tasks, reaching **0.914 macro-F1** (state-of-the-art on the benchmark).

X-Blocks — Linguistic Knowledge & VLM Generation Framework

[Code](#) | [Zenodo DOI](#) | [arXiv](#)

- Built an NLP analysis framework using **spaCy** dependency parsing and statistical log-odds ratios to surface scenario-specific linguistic patterns across **16,392 human-written driving explanations**.
- Developed a multi-LLM consensus annotation pipeline with automated tie-breaking via **GPT-4o**, validated against a human-expert gold standard ($n = 1,640$) at **91.45% agreement** ($\kappa = 0.91$).
- Evaluated zero-shot generation across **five frontier VLMs** on 476 driving clips using semantic, structural, and lexical metrics (**BLEU**, **ROUGE**, **METEOR**, **BERTScore**), showing significant gains in structural fidelity and vocabulary coverage ($p < 0.001$, paired Wilcoxon).

Teaching, Industry & Service

Tutor — Computing & Data for Engineers

Brisbane, QLD

Queensland University of Technology & QUT College

Feb 2024 – Present

- Instruct practical laboratory sessions covering advanced **Python** programming, data structures, algorithm design, software testing principles, and quantitative analysis.

AI Intern → Python Software Developer

Iran

HeyvaAI

Nov 2022 – Mar 2023

- Built core backend services for a financial social-trading platform; developed data ingestion scrapers, authentication flows, and high-performance REST APIs backed by **PostgreSQL**, **MongoDB**, **Redis**, and **Elasticsearch**.

Research Experience

Doctoral Researcher

Jun 2023 – Present

ARC Training Centre for Automated Vehicles in Rural & Remote Regions, QUT

Brisbane, QLD

- Direct primary research initiatives focused on multi-modal, human-centric reasoning architectures for automated driving systems, culminating in a first-author paper accepted to the **European Conference on Computer Vision (ECCV 2026)** alongside multiple peer-reviewed contributions.
- Designed and executed human-subject studies — real-world field trials and questionnaire instruments — for data collection and end-user trust evaluation of vision-language driving-explanation systems.
- Engineered an end-to-end data acquisition system utilizing **Python**, **PyQt**, and **ROS2** to interface with multi-camera arrays, LiDAR, and GNSS/IMU hardware, automating event logging and spatial-temporal alignment.

Selected Publications

- 2026** NARRATE: A Multimodal Real-World Australian Driving Dataset for Human-Centred Explanations in Automated Driving. Yousefi Zadeh, A., Zhu, Z., Li, X., Rakotonirainy, A., Glaser, S., Schroeter, R., Delhomme, P., & Mehraban, Z. **The 19th European Conference on Computer Vision (ECCV 2026)**, DriveX Workshop — Accepted, Archival Track. [\[OpenReview\]](#)
- 2026** X-Blocks: A Linguistic Knowledge Acquisition Framework for Scenario-Aware Explanations in Automated Vehicles. Yousefi Zadeh, A., Li, X., Rakotonirainy, A., Schroeter, R., Glaser, S., & Zhu, Z. **Advanced Engineering Informatics**, Elsevier — Under Review. [\[arXiv\]](#)
- 2026** Explainability in Automated Driving: From Spatial Attention to Human-Centred Reasoning. Rakotonirainy, A., Yousefi Zadeh, A., Zhu, Z., Benrachou, D., Elhenawy, M., Glaser, S., Li, X., Schroeter, R., Gouillou, M., & Delhomme, P. **Human Interaction and Emerging Technologies (IHET 2026)**. [\[DOI\]](#)

2025

PsyLingXAV: A Psycholinguistics Design Framework for XAI in Automated Vehicles. Yousefi Zadeh, A., Li, X., Rakotonirainy, A., Schroeter, R., & Glaser, S. [World Conference on eXplainable Artificial Intelligence \(xAI 2025\)](#), Istanbul, pp. 105–112, CEUR-WS. [\[Paper\]](#)

2024

Integrated Intelligent Control Systems for Eco and Safe Driving in Autonomous Vehicles. Yousefi Zadeh, A., Jamali, A., Mallipeddi, R., & Khayyam, H. [IEEE Transactions on Intelligent Transportation Systems](#), 25(12), 19444–19456. [\[DOI\]](#)

2024

Fuzzy Adaptive Cruise Control with Model Predictive Control for Automated Driving. Mehraban, Z., Yousefi Zadeh, A., Khayyam, H., Mallipeddi, R., & Jamali, A. [Engineering Applications of Artificial Intelligence](#), 136, 109008. [\[DOI\]](#)

Education

PhD, Computer Science

2023 – Present

[Queensland University of Technology \(QUT\)](#)

Brisbane, QLD

- Thesis: AI Model for Generating Human-Centric Explanations for Automated Vehicles.

MSc, Mechanical Engineering (Dynamics & Control Systems)

2019 – 2022

[University of Guilan, Iran](#)

BSc, Mechanical Engineering

2013 – 2018

[Islamic Azad University, Iran](#)

Awards, Service & Languages

Honors & Awards

- IEEE Human-Centric AI Summer School Scholarship, Swinburne University of Technology (2024).
- ARC Postgraduate Research Stipend & QUT Postgraduate Research Award (2023–2024).
- Runner-Up, QUT Visualise Your Thesis competition (2023).

Professional Service

- Technical Committee Member, IEEE Consumer Technology Society (2025–Present).
- Ad Hoc Committee Member, IEEE Vehicular Technology Society (2024–Present).
- Peer Reviewer, IEEE Transactions on Intelligent Transportation Systems & IEEE Access (2022–Present).

Languages

- English — Professional Working Proficiency; Persian — Native.